# The University of Azad Jammu and Kashmir, Muzaffarabad

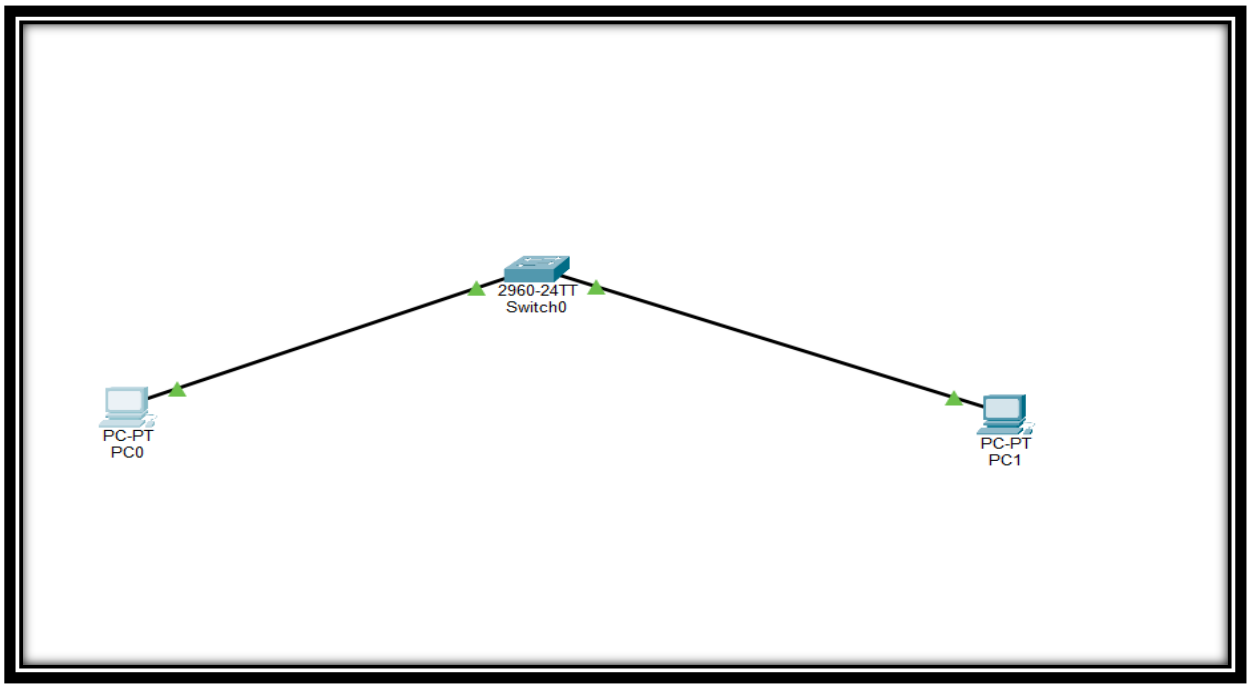
| Name | Danial Ayub Mughal |
|---|---|
| Roll Number | 2024-SE-20 |
| Class | 4rd Semester |
| Subject | Computer Networks (Lab task 01&02) |
| Course Code | CS-2203 |
| Submitted To | Engr. Dr .Asma Javaid |

**Department of Software Engineering**

## *Lab 01 & 02*

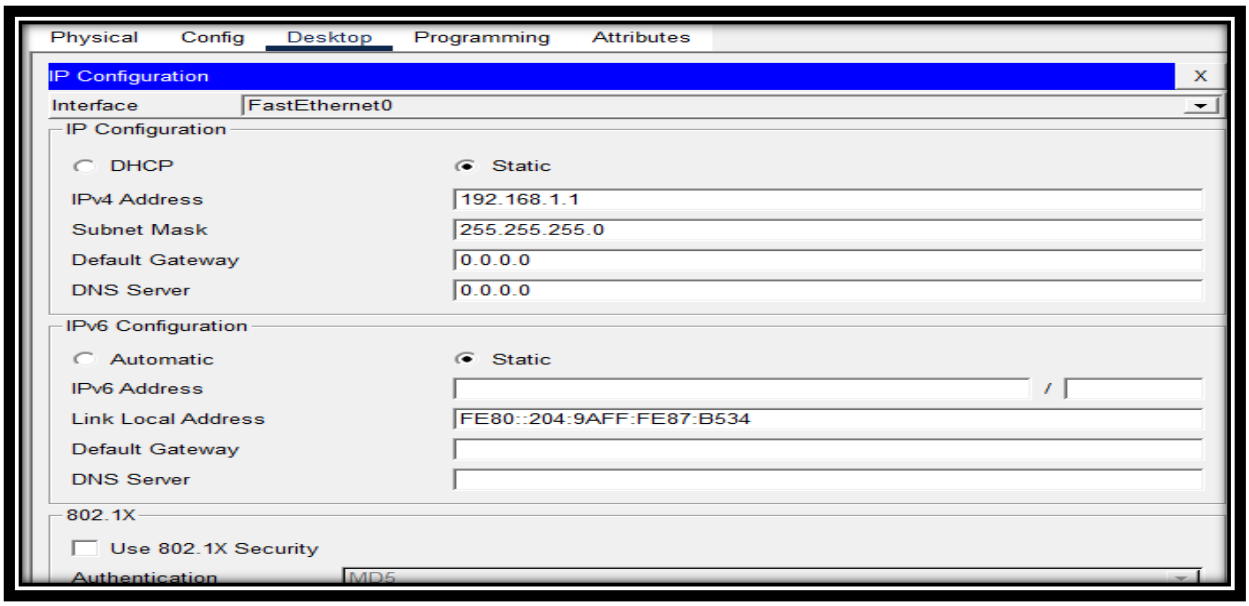
**Task: Create and test a simple LAN in Cisco Pocket Tracer**

**Part A-Network Topology Creation**

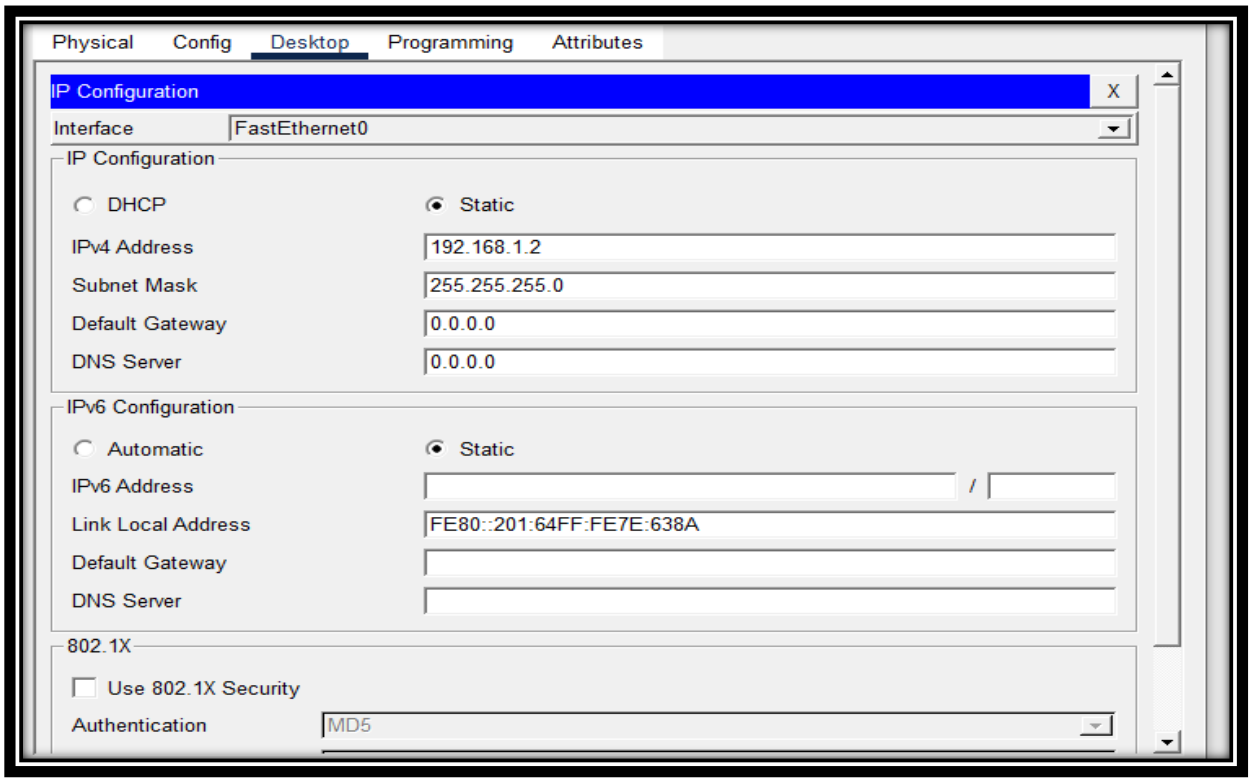



**Part B-IP Address Configuration**

- **PC0**



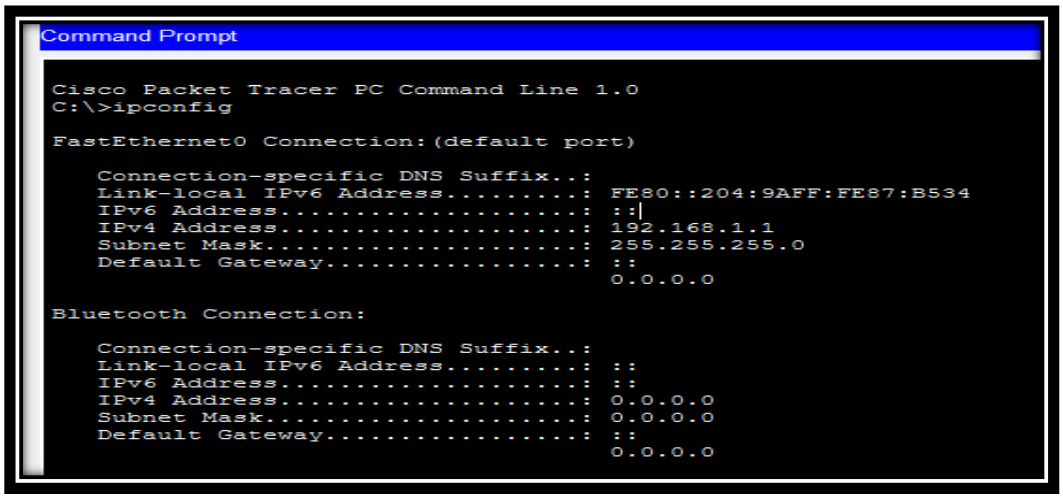
❖ **PC1**




## Part C- Network Testing Using CLI Commands

➢ **<u>For PC0</u>**

### 1. ipconfig



## 2. ping

From PC0:

Ping 192.168.1.2

```
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time=2ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 2ms, Average = 0ms
```

## 3. tracert

```
C:\>tracert
Cisco Packet Tracer PC Tracert

Usage: tracert target

C:\>tracert 192.168.1.2

Tracing route to 192.168.1.2 over a maximum of 30 hops:

  1   0 ms      0 ms      0 ms      192.168.1.2

Trace complete.
```

## 4. arp –a

```
C:\>arp -a
  Internet Address      Physical Address      Type
  192.168.1.2           0000.0cde.ccd9        dynamic
```

## 5. netstat

```
C:\>netstat

Active Connections

  Proto  Local Address          Foreign Address        State
```

## ➢ For PC1

### 1. ipconfig

```
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig

FastEthernet0 Connection:(default port)

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: FE80::200:CFF:FEDE:CCD9
   IPv6 Address....................: ::
   IPv4 Address....................: 192.168.1.2
   Subnet Mask.....................: 255.255.255.0
   Default Gateway.................: ::
                                     0.0.0.0

Bluetooth Connection:

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: ::
   IPv6 Address....................: ::
   IPv4 Address....................: 0.0.0.0
   Subnet Mask.....................: 0.0.0.0
   Default Gateway.................: ::
                                     0.0.0.0
```

### 2. ping

From PC1:

Ping 192.168.1.1

```
C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

## 3. tracert

```
C:\>tracert
Cisco Packet Tracer PC Tracert

Usage: tracert target

C:\>tracert 192.168.1.1

Tracing route to 192.168.1.1 over a maximum of 30 hops:

  1   0 ms      0 ms      0 ms      192.168.1.1
```

## 4. arp –a

```
C:\>arp -a
  Internet Address      Physical Address      Type
  192.168.1.1           0090.0c7e.e240        dynamic
```

## 5. netstat

```
C:\>netstat

Active Connections

  Proto  Local Address          Foreign Address        State
```